

**Centre for Development of Advanced Computing-
Noida**

Master of Computer Applications (MCA)

2nd Semester

RESEARCH PAPER

ON

**AcadXpert: Academic System for Digitalized
Management**

Group Members:

Stuti (09311804424)

Abhishek Pandey (10011804424)

Diksha Kumari (11711804424)

Ritesh Chaudhary (10411804424)

Ishan Jain (07611804424)

Laxman Sah (08611804424)

Under the guidance of

Mr. Sanjay Ojha

Abstract

Many colleges and universities are now using online systems to manage student admissions. However, most of the older systems have several problems. They often lack important features, are hard to use, and do not help students or admins manage documents or check the progress of applications properly. This research paper reviews these earlier systems and introduces an improved Admission Management System designed for postgraduate courses like MBA, MCA, and MTech. Our system makes the entire process smoother—from student registration and course selection to document uploads and admin review. It also allows students to check which documents are missing, update their forms, and see updates directly from the login page. The admin panel lets administrators easily filter and download applications based on courses. Compared to previous systems, our version is more user-friendly, better organized, and helps both students and administrators save time. This paper shows how the new system solves many of the problems found in older systems and offers a better way for colleges to handle admissions online.

1. Introduction

1.1 Background

In recent years, educational institutions have started using online systems to handle student admissions. This move toward digital platforms helps reduce paperwork, avoids long queues, and speeds up the admission process. Despite this shift, many existing systems still face problems like missing features, poor design, and lack of document tracking. These issues create confusion for students and make it difficult for administrators to manage applications efficiently.

1.2 Problem Statement

Most of the older admission systems do not allow students to track missing documents, update submitted forms, or see real-time updates from the admin side. Admins also face challenges in filtering and downloading student data based on selected courses. These limitations slow down the admission process and lead to errors or delays. In the field of MCA/IT, solving such system-level problems is important because it helps improve automation, database handling, and user experience design—all of which are key areas in modern software development.

1.3 Objectives

Primary Objective:

To develop an improved web-based admission system that simplifies the admission process for postgraduate courses like MBA, MCA, and MTech.

Secondary Objectives:

- To allow students to track and update their application and document status.
- To enable administrators to filter, download, and manage application data based on course and semester.
- To compare this system with previous ones and highlight the improvements made.

1.4 Scope and Limitations

Scope:

- The system focuses on managing student admissions for postgraduate programs.
- It includes modules for student registration, document uploads, application tracking, and an admin panel.
- Designed for use in colleges and universities looking to digitize their admission process.

Limitations:

- The system does not include features like payment integration or email/SMS notification.
- Real-time communication between admin and student is not currently supported.
- The system's performance depends on the quality of internet connectivity and server setup.

2. Literature Review

Over the past 5–7 years, many educational institutions have started adopting online admission systems to replace manual paper-based processes. Researchers and developers have explored

various methods to improve these systems, such as secure login mechanisms, cloud-based storage for student records, and basic dashboards for administrative users.

In a 2018 study by Sharma et al., a basic student registration system was introduced using PHP and MySQL. While it allowed students to apply online, it lacked document tracking and did not support admin filtering options. Similarly, a 2019 project developed using Java focused on data entry and report generation but didn't allow students to update their forms or get real-time feedback on missing documents.

Another 2020 study implemented a cloud-based document management system using Firebase. This version improved storage reliability but still failed to include role-based filtering for administrators and proper UI for students. In 2021, researchers used Google Forms and Sheets for small colleges to collect admission data. Although cost-effective, this solution lacked security, automation, and scalability.

Some more advanced systems introduced login authentication and used cloud technologies like AWS or Firebase to store application data. However, these systems were either too complex for smaller institutions or missed important user-centered features like form editing, document alerts, or course-based filtering in admin dashboards.

Gaps Identified in Previous Systems

- Most systems lacked the option for students to update or track their application status after submission.
- Admin panels were often basic, with no filters to sort applications by course or semester.
- No real-time pending document alerts were available.
- Poor user interfaces led to confusion for both students and staff.
- Many systems were not mobile-friendly or lacked responsive design.

Contribution of the Current System

This research addresses the above gaps by introducing:

- A complete form review and update feature for students.
- A course- and semester-based filter for admins.

- A “pending documents” alert displayed during login.
- A clean, easy-to-use interface for both students and administrators.

By reviewing these existing systems and identifying their limitations, this study presents a more practical, user-focused solution suitable for institutions of various sizes.

3. Methodology

3.1 Research Design

- **Approach:**
 - This project follows a **Qualitative Design** approach.
 - Focuses on reviewing existing systems and identifying their flaws to propose a better solution.
 - The system was evaluated through **comparative analysis** rather than statistical modelling.
- **Experimental Setup:**
 - A web-based prototype was built and tested for user interface, data accuracy, and ease of admin operations.
 - Simulated student data and test cases were used to mimic real-time usage.

3.2 Tools & Technologies Used

- **Frontend:** React.js, HTML5, CSS3, JavaScript
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Tools:** VS Code, Postman, npm
- **Hosting:** Localhost (Dev), Vercel & Render (Prod)

3.4 Data Collection & Processing

- **Data Source:**

- Dummy student data was created manually for testing purposes.
- Real college admission data was **not used** due to privacy constraints.
- **Data Processing Steps:**
 - Input forms validated using client-side and server-side scripting.
 - Uploaded documents were stored in structured folders and linked to database records.
 - Admin panel accessed the data through secure queries to retrieve, filter, and manage records.

3.5 Implementation Steps

1. Requirement Analysis:

- Studied existing systems and identified their drawbacks.

2. Database Design:

- Created tables for users, courses, semesters, documents, and admin login.

3. Frontend Development:

- Designed user interface for registration, login, form submission, and status viewing.

4. Backend Logic:

- Developed Node.js and Express.js APIs to handle form submissions, application updates, and admin-side filtering functionalities.

5. Admin Dashboard:

- Created a separate module where admins could view and download filtered student applications.

6. Testing & Debugging:

- Ran the system with test data to fix bugs, improve UI/UX, and ensure proper data flow.

7. Final Review:

- Compared the final system with previous ones and confirmed improved performance and usability.

4. Results & Discussion

4.1 Experimental Results

The project was tested using sample student records. The goal was to check how well the system performs in real-life situations. Below are the results of the key features tested:

Feature Tested	Result	Remarks
Student Registration & Login	Working	Easy to use and responsive
Form Submission & Update	Working	Students could submit and edit forms
Document Upload	Working	Documents were saved correctly
Pending Document Alert	Working	Shown to students during login
Admin Filtering (by Course/Semester)	Working	Admins could filter data easily
Data Download (PDF/Excel)	Working	Reports were downloaded successfully

Feedback from Test Users:

- Most students found the system easy to use.
- Admin users appreciated the filtering and data export features.
- The system responded quickly even when tested with a considerable volume of data.

4.2 Comparative Analysis

This project was compared with other earlier admission systems. Here's a summary of how this project performs better:

Feature	Old Systems	This Project
Form Editing	Not available	Students can update forms
Document Status Alert	Not shown	Visible during student login
Admin Filtering	Very limited or not available	Admins can filter by course/semester
User Interface	Outdated and hard to use	Simple and modern using React.js
Performance	Slows down with more data	Fast due to Node.js and Express
Scalability	Mostly limited to local systems	Can be deployed online easily

4.3 Interpretation

- **Why the system worked well:**
 - React.js made the interface fast and easy to use.
 - Node.js and Express made the backend efficient.
 - The system was built in a modular way, making it easy to test and improve.
- **Issues faced:**
 - File upload security needed extra steps.
 - Some minor errors occurred when updating forms, but were fixed during testing.
- **What can be improved in the future:**
 - Add email or SMS alerts to notify students.
 - Create separate admin roles for better control.
 - Add a dashboard showing application statistics.
 - Option to integrate online payment for application fees.

5. Conclusion & Future Work

5.1 Conclusion

This project aimed to improve the online admission process by developing a web-based system using React.js and Node.js. The system allows students to register, upload documents, and update their applications, while giving administrators the ability to filter and manage records easily.

All the main objectives of the project were successfully achieved:

- Students could submit and update forms with required documents.
- Admins were able to filter applications by course and semester.
- The system was tested and found to be user-friendly and fast.

Compared to older systems, this project provided better performance, flexibility, and a more modern user experience.

5.2 Future Enhancements

To make the system more powerful and useful, the following improvements can be added in the future:

- **Email or SMS notifications** to inform students about missing documents or application status.
- **Admin role management** to allow multiple users with different access levels.
- **Dashboard analytics** to view reports such as the number of applications received, approved, or pending.
- **Online payment integration** for admission or registration fees.
- **Mobile app version** for easier access on smartphones.
- **Stronger security features** like OTP verification or data encryption.

6. References

1. Aggarwal, A., & Bhatia, R. (2020). *Modern Web Development with React and Node*. Packt Publishing.
2. Kumar, R., & Sharma, P. (2021). "Enhancing Online Admission Systems Using MERN Stack." *International Journal of Computer Applications*, 175(3), 12–18.
3. Gupta, N. (2019). "Automation of College Admission Process: A Review." *International Journal of Engineering Research and Applications*, 9(5), 66–70.
4. Singh, M., & Yadav, S. (2020). "User-Centric Design in Web-Based Education Portals." *Journal of Web Engineering*, 18(4), 295–310.
5. Wieruch, R. (2022). *The Road to React: Your Journey to Master React.js*. Independently published.
6. Osmani, A. (2018). *Learning JavaScript Design Patterns*. O'Reilly Media.
7. Raj, D., & Iyer, V. (2021). "Comparative Study on Traditional and Web-Based Admission Management Systems." *ICTACT Journal on Soft Computing*, 11(3), 1234–1240.
8. Sahu, A. K. (2020). *Full Stack Development with MERN*. BPB Publications.
9. Sharma, M., & Thakur, A. (2020). "Role of Online Portals in Simplifying Student Admission Process." *International Journal of Information Technology*, 6(1), 21–27.
10. MongoDB Inc. (2023). *MongoDB Documentation*. <https://www.mongodb.com/docs/>
11. Express.js Team. (2023). *Express.js Guide*. <https://expressjs.com/>
12. Meta (Facebook). (2023). *React – A JavaScript Library for Building User Interfaces*. <https://reactjs.org/>
13. Node.js Foundation. (2023). *Node.js Documentation*. <https://nodejs.org/en/docs/>
14. Ahmad, T., & Khan, F. (2021). "Improving Student Data Management Using Web Technologies." *International Journal of Computer Science Trends and Technology*, 9(2), 33–40.
15. Jain, R. (2020). "Digital Transformation in College Admission Procedures." *International Journal of Advanced Research in Computer Science*, 11(7), 223–229.

16. Roy, A., & Sen, D. (2022). "Secure Online Form Submission using Web Technologies." *International Journal of Computer and Communication Engineering*, 8(5), 101–108.
17. World Wide Web Consortium (W3C). (2022). *HTML and Web Standards*.
<https://www.w3.org/>
18. Stack Overflow Developer Survey. (2023). *Most Popular Web Technologies*.
<https://survey.stackoverflow.co/2023/>